

ECA-RAG: Calibrated Confidence, Hop-Conditioned Re-Ranking, and the Limits of Adaptive- k Retrieval in Multi-Hop RAG

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Abstract—Adaptive retrieval-augmented generation modulates passage depth k to balance sufficiency against overhead. We evaluate calibrated confidence gating and iterative adaptive- k policies on HotpotQA ($N = 2640$ matched-depth). First, cross-encoder logits are miscalibrated; Platt scaling reduces Expected Calibration Error from 0.225 to 0.039 and Brier score from 0.213 to 0.116 across 5,385 held-out chunk evaluations ($N = 540$). Second, calibrated gating fails to surpass fixed- k at matched depth $k = 4.00$: single-pass gating trails fixed- k by 2.7 points ($\Delta = -0.027$, Holm $p < 0.001$), while iterative gated traversal does not differ significantly from fixed- k ($\Delta = -0.0088$, raw $p = 0.079$, Holm $p = 0.079$), losing 2.6 points against ungated iteration ($\Delta = -0.0257$, Holm $p < 0.001$). Third, sequential hop-conditioned reformulation brings modest bridge gains (+1.7 points, $\Delta = +0.017$, Holm $p = 0.001$) while degrading comparison queries (−2.4 points, $\Delta = -0.024$, Holm $p = 0.007$). Finally, an exploratory multi-depth sweep demonstrates that a supervised sufficiency stopper (ROC AUC 0.80) matches fixed- k on bridge queries across $k \in [2.0, 3.5]$, yet degrades comparison performance ($\Delta = -0.138$ to -0.044 , all Holm $p < 0.001$), while outpacing iterative gating on both question types (all Holm $p < 0.001$).

Index Terms—Information retrieval, multi-hop reasoning, confidence calibration, adaptive passage depth, cross-attention re-ranking.

I. Introduction

Grounding autoregressive language models upon external knowledge indices—a paradigm widely formalized as Retrieval-Augmented Generation (RAG)—serves as a primary bulwark against factual hallucinations while incorporating timely real-world information [1], [2]. Conventional RAG pipelines enforce an invariant retrieval budget, extracting an identical quantity (k) of candidate passages for every query without regard to task complexity or source ambiguity [3], [4]. In multi-hop question answering, where deducing a correct resolution requires chaining disconnected evidence spans distributed across distinct documents [5], static context limits induce an acute tension. Small values of k risk omitting vital bridging assertions (under-retrieval), whereas aggressive context expansion introduces distracting irrelevancies that degrade generator attention while inflating inference compute (over-retrieval).

To circumvent this rigid dilemma, an emerging body of research explores dynamic mechanisms capable of regulating context acquisition on the fly [3], [4], [6], [7]. A standard strategy employs confidence-governed gating, retaining only candidate text passages whose relevance estimates exceed a predefined decision threshold [1]. Nonetheless, deep transformer architectures and cross-encoder re-rankers are notoriously miscalibrated, often exhibiting substantial probability overconfidence [8]. Because raw logit outputs cannot be interpreted as well-behaved posterior probabilities of relevance, heuristic thresholding rules remain brittle and vulnerable to distributional divergence between query categories.

In this work, we conduct a comprehensive empirical appraisal examining the joint behavior of post-hoc probability recalibration, hop-conditioned query reformulation, and adaptive- k stopping criteria within multi-hop RAG. We analyze six retrieval mechanisms on HotpotQA [5], scaling evaluation from an initial calibration partition to an extensive pooled sample of $N = 2640$ question–document instances.

Crucially, adaptive stopping rules inevitably alter the average context budget (k) alongside retrieval recall, meaning uncalibrated comparisons easily confound algorithmic selection capability with the statistical advantage of inspecting more text. To isolate genuine algorithmic performance from raw passage volume, our evaluation enforces rigorous matched-depth benchmarking fixed at reference depth $k = 4.00$, complemented by an exploratory evaluation over target depths $k \in \{2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5\}$ using 10,000 paired bootstrap resamples.

Our primary empirical findings are:

- 1) **Calibration Quality:** Raw cross-encoder relevance logits exhibit substantial miscalibration on the held-out $N = 540$ evaluation split (ECE = 0.2248, Brier score 0.2127, evaluated over the exact 5,385 candidate chunk predictions from the 540-example held-out split, not the $N = 2640$ pool). Chunk-level Platt scaling [9] reduces ECE by 82.6% to 0.0391 and cuts Brier score to 0.1157.

- 2) Limits of Calibrated Gating at $k = 4.00$: Gating retrieval on calibrated chunk probabilities does not exceed static fixed- k baselines at matched depth $k = 4.00$. Calibrated single-pass gating underperforms fixed- k re-ranking by 2.7 points recall on bridge questions ($\Delta = -0.0266$, 95% CI $[-0.0341, -0.0191]$, Holm $p < 0.001$). Iterative gated retrieval yields recall statistically indistinguishable from fixed- k on bridge questions ($\Delta = -0.0088$, 95% CI $[-0.0187, +0.0010]$, raw $p = 0.0794$, Holm $p = 0.0794$ within the 5-contrast bridge family at $k = 4.00$, 95% CI includes zero), but costs 2.6 points recall relative to ungated iteration ($\Delta = -0.0257$, 95% CI $[-0.0328, -0.0186]$, Holm $p < 0.001$).
- 3) Asymmetric Multi-Hop Reformulation: Hop-conditioned query reformulation at fixed depth $k = 4.00$ yields a modest, statistically significant gain on bridge questions (+1.7 points, $\Delta = +0.017$, 95% CI $[+0.0077, +0.0258]$, Holm $p = 0.001$), but degrades comparison questions (-2.4 points, $\Delta = -0.024$, 95% CI $[-0.0391, -0.0089]$, Holm $p = 0.007$).
- 4) Depth-Dependent Trajectories and Sufficiency Stopping: In exploratory multi-depth analysis, iterative gating lags behind fixed- k at tight budgets ($k \leq 3.5$, all Holm $p \leq 0.005$) and converges with fixed- k from $k = 4.0$ to $k = 5.0$. At $k = 5.5$, it attains an exploratory +0.0127 recall (95% CI $[+0.0044, +0.0210]$, Holm $p = 0.0176$). A learned sufficiency stopping classifier (ROC AUC 0.7964) remains statistically indistinguishable from fixed- k on bridge questions across all covered depths ($k \in [2.0, 3.5]$, with 95% CIs spanning zero). However, on comparison questions, the stopper is significantly worse than fixed- k at every covered depth $k \in [2.0, 3.5]$ ($\Delta = -0.138$ to -0.044 , all Holm $p < 0.001$). Nonetheless, in exploratory analysis it improves upon iterative gating (Arm 5) on both question types at $k \in [2.0, 3.5]$ (+1.5 to +5.3 points on bridge, +3.4 to +5.5 points on comparison, all Holm $p < 0.001$).

II. Related Work

A. Context-Aware and Re-Ranked RAG

Traditional dense passage retrieval systems rely upon dual-encoder architectures that project queries and candidate documents into vector representations asynchronously, facilitating rapid sub-linear search while forfeiting fine-grained token-level cross-attention. Incorporating cross-encoder re-ranking stages substantially sharpens precision by computing bidirectional cross-attention over fused query-document sequences. Collini et al. [1] formulated Context-Aware RAG (CA-RAG), showing that pruning semantic incongruities through similarity validation enhances downstream factual alignment. Within our evaluation taxonomy, Arm 1 serves as a CA-RAG-aligned dense bi-encoder benchmark, upon which we sys-

tematically contrast cross-encoder re-ranking, parametric calibration, and multi-hop gating policies.

B. Adaptive and Dynamic Retrieval

Adaptive retrieval paradigms modulate the frequency or depth of corpus interaction according to query-specific indicators. Kratzwald and Feuerriegel [3] explored query-adaptive retrieval horizons for complex question answering pipelines. Adaptive-RAG [4] dynamically steers requests across non-retrieval, single-step, or iterative pathways utilizing a dedicated query difficulty classifier. Training-free gating strategies, exemplified by TARG [6] (which decides retrieval necessity via draft-token entropy and logit margin) and CtrlA [7] (which steers retrieval through honesty and confidence directions in LLM internal representations), evaluate when context lookup is justified. For extended generative synthesis, proactive architectures such as FLARE [10], Self-RAG [2], and DRAGIN [11] activate retrieval when internal token certainty drops below specified thresholds. Nevertheless, existing frameworks predominantly emphasize the binary trigger of retrieval rather than resolving the optimal passage count (k) under strict matched-depth controls.

C. Confidence Calibration in Neural Systems

Guo et al. [8] established that deep neural models systematically output overconfident probability estimates that misstate true empirical correctness. Post-processing recalibration techniques, notably temperature scaling and Platt scaling [9], optimize parametric functions over held-out calibration partitions to transform uncalibrated raw scores into faithful posterior probabilities. In this paper, we examine whether recalibrating cross-encoder relevance logits can provide dependable stopping boundaries during multi-hop document retrieval.

D. Multi-Hop Retrieval Architectures

Multi-hop question answering necessitates tracing multi-step reasoning trajectories across disjoint passages. Architectures such as Multi-hop Dense Retrieval (MDR) [12] and Baleen [13] train specialized multi-hop encoders to navigate dense document networks. Synergistic approaches like IRCOT [14] alternate chain-of-thought token generation with intermediate retrieval steps. Rather than training specialized multi-hop representations from scratch, our work evaluates whether post-hoc probability calibration and confidence-based halting can effectively manage retrieval depth using widely deployed, off-the-shelf bi-encoders and cross-encoders.

III. Methodology

A. System Architecture and Evaluation Arms

We benchmark six distinct retrieval methodologies (Arms 1–6) operating over the standard candidate pool of ~ 10 passages per HotpotQA item in the distractor configuration:

- Arm 1 (Dense Baseline): Encodes query q and retrieves the top- k passages ranked by cosine similarity using a bi-encoder: $\mathcal{D}_1 = \text{Top-}k_{d \in \mathcal{P}} \langle \mathbf{e}_q, \mathbf{e}_d \rangle$, where $\mathcal{P}$ is the per-question candidate pool.
- Arm 2 (Fixed- k Cross-Encoder): Retrieves an initial candidate pool of $M = 10$ passages using Arm 1, then computes full cross-attention relevance logits $s(q, d)$ with a cross-encoder (truncated to 512 tokens). Passages are selected under fixed integer cutoffs $k \in \{1, 2, 3, 4, 5, 6\}$.
- Arm 3 (Uncalibrated Marginal Gating): Traverses the ranked passage list generated by the cross-encoder, retaining candidates whose unnormalized logit satisfies $s(q, d) \geq s_{\text{thresh}}$, bounded between a minimum of 1 and a maximum of 6 chunks.
- Arm 4 (Calibrated Single-Pass Gating): Maps raw logits s into calibrated probabilities $\hat{p} = \sigma(a \cdot s + b)$ via Platt scaling and steps through the ordered list, selecting items meeting $\hat{p} \geq \theta$, constrained between 1 and 6 chunks. We examine both global pooled scaling (Arm 4a) and question-type-conditioned parameters (Arm 4b).
- Arm 5 (Calibrated Iterative Adaptive- k): Performs iterative multi-hop traversal up to 6 hops over the per-question pool. At hop 0, remaining passages are evaluated against query q alone; at each successive hop $h \geq 1$, the cross-encoder re-evaluates all remaining candidates against q concatenated with the full text of the most recently retrieved chunk ($q \oplus \text{“ ”} \oplus d_{(h)}$). The top candidate is chosen; for hops $h \geq 1$, traversal terminates whenever its per-type Platt probability falls below θ (max 6 chunks). No bi-encoder is re-invoked once the initial candidate set is formed; a 50-word prefix truncation baseline is analyzed strictly as an ablation in Section IV-G.
- Arm 6 (Learned Sufficiency Stopping): Implements a supervised logistic regression classifier with standardized feature preprocessing trained on calibration traces. The model leverages 6 features: calibrated probability of the incoming candidate, its raw logit, hop index, probability differential from the preceding hop, minimum calibrated probability among already selected passages, and a binary indicator for bridge questions (`is_bridge`). The binary target signals whether all gold supporting passages have been accumulated. At each hop $h \geq 1$, retrieval halts if candidate confidence drops below θ or if the posterior $P(\text{sufficient} \mid \mathbf{x}) \geq \tau$ (subject to a 6-chunk maximum).

B. Chunk-Level Platt Calibration

Given a cross-encoder assigning raw continuous scores $(q, d) \rightarrow s \in \mathbb{R}$, we fit an affine logit adjustment passed through the standard logistic sigmoid function [9]:

$$P(y = 1 \mid s) = \sigma(a \cdot s + b) = \frac{1}{1 + \exp(-(a \cdot s + b))} \quad (1)$$

where $y \in \{0, 1\}$ represents ground-truth supporting document relevance. Parameters $a, b \in \mathbb{R}$ are determined on the calibration set by minimizing binary cross-entropy loss:

$$\mathcal{L}(a, b) = - \sum_{i=1}^{N_{\text{cal}}} [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)] \quad (2)$$

Calibration fidelity is measured via Expected Calibration Error (ECE) across $M = 10$ uniform probability bins $B_m \subset (0, 1]$:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (3)$$

paired with the mean squared probability error (Brier score): $\text{BS} = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - y_i)^2$.

C. Hop-Conditioned Iterative Re-Ranking

In multi-hop settings, early search steps often retrieve an introductory passage whose textual body uncovers bridging entities necessary for subsequent inference. In Arm 5 and ungated multi-step retrieval, traversal iterates up to 6 hops over the per-query pool of ~ 10 candidates without re-executing bi-encoder retrieval. At hop 0, candidate texts are matched against q alone. At each subsequent hop $h \geq 1$, the cross-encoder re-ranks all unselected candidates against the prompt augmented with the complete passage selected at hop $h - 1$:

$$q^{(h+1)} = q \oplus \text{“ ”} \oplus d_{(h)} \quad (4)$$

The highest-ranking available candidate is extracted. In Arm 5, for hops $h \geq 1$, search halts if the candidate’s type-specific Platt probability drops below θ (min 1, max 6 chunks). In Section IV-G, we contrast this complete passage concatenation with a 50-word prefix truncation variant ($q \oplus \text{“ ”} \oplus \text{Trunc}_{50}(d_{(h)})$) to verify whether excessive passage length induces token dilution during cross-attention re-ranking.

D. Reproducibility Protocol and Frozen Parameters

To ensure reproducibility:

- Dataset and Splits: We use the HotpotQA validation set in the distractor setting ($N = 3000$ total examples drawn from the full validation set of 7,405 examples, each paired with a candidate pool of ~ 10 paragraphs) [5]. The 3,000 examples are the first 3,000 of the seed-42 shuffled HotpotQA distractor validation set. A seed-42 random split partitioned the first 900 example IDs into the first 40% ($N = 360$; 279 bridge, 81 comparison) for calibration and the remaining 60% ($N = 540$; 424 bridge, 116 comparison) for held-out evaluation (these are not contiguous index slices of the unshuffled dataset). The extension split ($N = 2100$; 1653 bridge, 447 comparison) comprises elements 900 to 2999 from the same seed-42 shuffled HotpotQA distractor validation set; its example IDs

TABLE I
Cross-Encoder Calibration Quality on $N = 540$ Evaluation Split
(5,385 Candidate Chunk Predictions)

Calibration Method	ECE ↓	Brier Score ↓	ECE Red.
Raw Sigmoid $\sigma(s)$	0.2248	0.2127	Baseline
Platt Scaling (Pooled)	0.0419	0.1161	81.4%
Platt Scaling (Per-Type)	0.0391	0.1157	82.6%

are verified to be strictly disjoint from the first 900 examples. The pooled dataset ($N = 2640$; 2077 bridge, 563 comparison) concatenates the evaluation and extension splits.

- Models and Versions: Dense retrieval uses sentence-transformers/all-MiniLM-L6-v2 (384-dimensional embeddings, cosine similarity). Cross-encoder re-ranking uses cross-encoder/ms-marco-MiniLM-L-6-v2 with a 512-token truncation limit.
- Frozen Calibration Parameters: Parameters fit on the $N = 360$ calibration split are held frozen across all evaluation splits:
 - Platt parameters: Pooled ($a = 0.2703, b = -1.3639$); Bridge ($a = 0.2454, b = -1.3621$); Comparison ($a = 0.4724, b = -1.4665$).
 - Confidence threshold: $\theta^* = 0.18$ was selected on the calibration split to maximize recall subject to mean depth $k \leq 4.0$. Sensitivity is evaluated at $\theta = 0.15$.
 - Uncalibrated threshold: $s_{\text{thresh}} = -1.9250$ (median logit on calibration split).
 - Sufficiency stopper: $\tau^* = 0.95$ (tuned on calibration traces).
- Random Seed: All data partitioning and bootstrap iterations use seed 42.

IV. Empirical Evaluation and Results

A. Calibration Quality

Table I reports calibration metrics for raw versus Platt-scaled cross-encoder logits on the held-out $N = 540$ evaluation set (exact 5,385 candidate chunk predictions, not the pooled $N = 2640$ cohort). Raw logits mapped through a standard sigmoid display high miscalibration (ECE = 0.2248, Brier score 0.2127), overestimating the probability of relevance for negative distractor paragraphs.

Platt scaling reduces ECE to 0.0419 (pooled) and 0.0391 (per-type), achieving an 82.6% relative reduction in calibration error. The Brier score drops by 45.6% to 0.1157. Conditioning calibration on question type provides a minor gain, reflecting slightly different logit distributions between bridge and comparison queries. Platt parameters transferred robustly to the $N = 2100$ extension split (20,901 candidate chunk predictions): raw logits exhibited ECE = 0.2143 and Brier score 0.2039, whereas frozen per-type Platt scaling achieved ECE = 0.0373 (an 82.6% relative reduction) and Brier score 0.1146 (pooled Platt:

ECE = 0.0389, Brier score 0.1162), demonstrating that the calibration parameters generalize stably across independent question sets.

B. Matched-Depth Evaluation at Reference Depth $k = 4.00$

Table II presents our matched-depth results at exact mean depth $k = 4.00$, comparing static fixed- k retrieval against ungated iterative retrieval and calibrated adaptive gating across the pooled $N = 2640$ cohort as well as the constituent $N = 540$ and $N = 2100$ splits.

1) Bridge Questions: On the pooled bridge cohort ($N = 2077$), ungated iterative retrieval attains a recall of 0.8240, compared to 0.8072 for fixed- k retrieval ($\Delta = +0.0169$, 95% CI $[+0.0077, +0.0258]$, $p < 0.001$, Holm $p = 0.001$). Gating retrieval on calibrated scores does not improve recall over fixed- k : Arm 5 attains an interpolated recall of 0.7983, displaying no statistically significant divergence from fixed- k ($\Delta = -0.0088$, 95% CI $[-0.0187, +0.0010]$, raw $p = 0.0794$, Holm $p = 0.0794$).

Relative to ungated iteration at identical depth, gating incurs a recall cost of 2.6 points ($\Delta = -0.0257$, 95% CI $[-0.0328, -0.0186]$, $p < 0.001$, Holm $p < 0.001$). Calibrated single-pass gating (Arm 4b) performs significantly below fixed- k at matched depth, reaching 0.7806 recall ($\Delta = -0.0266$, 95% CI $[-0.0341, -0.0191]$, $p < 0.001$).

2) Comparison Questions: On the pooled comparison cohort ($N = 563$), static fixed- k retrieval attains 0.9538 recall. Ungated iterative search attains a lower recall of 0.9298 ($\Delta = -0.0240$, 95% CI $[-0.0391, -0.0089]$, $p = 0.002$, Holm $p = 0.007$). Arm 4b calibrated single-pass gating achieves 0.9520 recall, which is indistinguishable from fixed- k ($\Delta = -0.0018$, 95% CI $[-0.0136, +0.0095]$, raw $p = 0.7202$, Holm $p = 0.7202$). Iterative gated retrieval (Arm 5) reaches 0.9147 ($\Delta = -0.0391$, 95% CI $[-0.0565, -0.0221]$, $p < 0.001$).

C. Pareto Frontiers of Retrieval Efficiency

Fig. 1 shows the empirical Pareto frontiers of Paragraph Recall versus Mean Retrieval Depth (k) across bridge and comparison subsets on the pooled $N = 2640$ dataset, displaying pointwise 95% bootstrap confidence bands ($B = 10,000$).

- Bridge Structure: The fixed- k curve traces a concave frontier across retrieval depths (0.6810 at $k = 2.0$ to 0.8533 at $k = 5.5$). Calibrated single-pass gating (Arm 4b) lies below the fixed- k envelope through $k = 5.0$. Iterative gating (Arm 5) shifts the adaptive frontier upward, converging with fixed- k at $k \geq 4.0$. At $k = 4$, ungated iterative retrieval clears the fixed- k curve, whereas at $k = 5$ it sits marginally above it.
- Comparison Structure: The fixed- k curve forms an upper envelope. All adaptive gating sweeps (Arm 4b, Arm 5, Arm 6) remain bounded by fixed- k . As gating becomes more aggressive (higher θ or τ), recall falls faster than under static top- k truncation.

TABLE II
Matched-Depth Paragraph Recall at Exact Mean Depth $k = 4.00$ Across Evaluation Cohorts

Split / Dataset	Type (N)	Fixed- k ($k = 4$)	Iterative ($k = 4$, No Gate)	Arm 5 (Interp. $k = 4$)	Arm 4b (Interp. $k = 4$)	Arm 6 (Interp. $k = 4$)
$N = 540$ Component Split	Bridge (424)	0.8054	0.8160	0.8013	0.7693	n/a
	Comparison (116)	0.9483	0.9181	0.8649	0.9479	0.8712*
$N = 2100$ Component Split	Bridge (1653)	0.8076	0.8261	0.7978	0.7835	n/a
	Comparison (447)	0.9553	0.9329	0.9273	0.9530	0.9280*
Pooled Headline ($N = 2640$)	Bridge (2077)	0.8072	0.8240	0.7983	0.7806	n/a
	Comparison (563)	0.9538	0.9298	0.9147	0.9520	0.9168*

Note on Linear Interpolation and Arm 6: To ensure rigorous comparisons at matched retrieval depth $k = 4.00$, points for Arm 5 and Arm 4b are evaluated via piece-wise linear interpolation between empirical sweep points. For Arm 5 on bridge, the bracketing points are $\theta = 0.18$ ($k = 3.8671$, recall 0.7913) and $\theta = 0.16$ ($k = 4.2297$, recall 0.8105), linearly interpolating to recall 0.7983 at $k = 4.0000$. Arm 6 values on bridge are reported as n/a because early sufficiency stopping caps its maximum depth near $k = 3.87$ under frozen calibration threshold $\theta^* = 0.18$, so it never reaches depth $k = 4.00$. On comparison questions, Arm 6 attains mean depths up to $k \approx 4.04$ – 4.20 , producing point estimates of 0.8712 ($N = 540$), 0.9280 ($N = 2100$), and 0.9168 (Pooled, marked *). However, because under 95% of paired bootstrap draws achieve $k = 4.00$, non-parametric hypothesis tests and confidence intervals cannot be reliably computed at this depth.

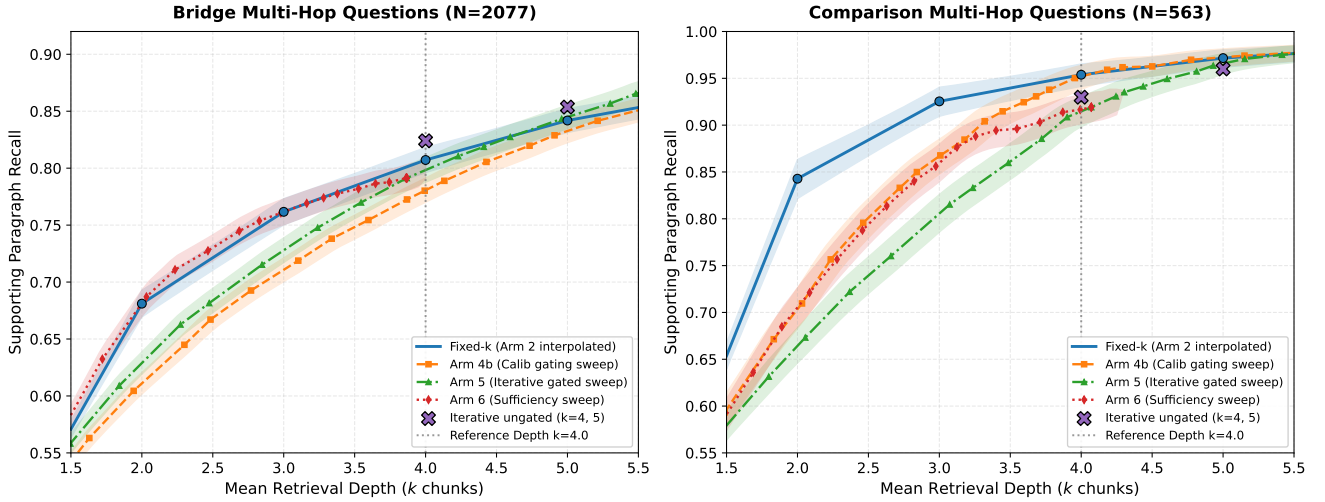


Fig. 1. Recall-efficiency Pareto frontiers across mean retrieval depth (k) on pooled HotpotQA validation questions ($N = 2640$; bridge $N = 2077$, comparison $N = 563$, pooled from the 540 held-out evaluation and 2100 extension splits). Solid blue curve denotes static Fixed- k retrieval (Arm 2), where the continuous line represents the linear mixture of $\lfloor k \rfloor$ and $\lceil k \rceil$, an achievable time-sharing baseline rather than a smoothing artifact, with discrete markers at integer depths $k \in \{1, 2, 3, 4, 5, 6\}$. Dashed orange curve represents Calibrated Single-Pass Gating (Arm 4b, swept over θ). Dash-dotted green curve represents Calibrated Iterative Gating (Arm 5, swept over θ). Dotted red curve denotes Learned Sufficiency Stopping (Arm 6, swept over τ with $\theta^* = 0.18$). On bridge questions Arm 6 does not reach $k \geq 4.0$ because its inner candidate threshold $\theta^* = 0.18$ caps its maximum depth near $k = 3.87$; on comparison it reaches k of about 4.04–4.20. Shaded ribbons represent pointwise 95% paired bootstrap confidence bands across $B = 10,000$ resamples (not simultaneous confidence bands; overlapping bands do not imply lack of statistical significance, as paired differences in Table V are tighter). Purple markers indicate Ungated Iterative retrieval at fixed depths $k = 4$ and $k = 5$. The vertical dotted line marks the $k = 4.0$ reference operating budget. Both panels use a common y-axis lower limit of 0.55 for direct visual comparability.

D. Full Evaluation Split Results

Table III presents retrieval metrics across both the initial $N = 540$ evaluation split and the independent $N = 2100$ extension split.

1) Impact of Cross-Encoder Re-Ranking: Across both splits, adding the cross-encoder re-ranker at fixed $k = 4$ (Arm 2 vs. Arm 1) improves bridge recall by +0.0495 on $N = 540$ ($p < 0.001$) and +0.0647 on $N = 2100$ ($p < 0.001$). Full-match accuracy increases from 0.5275 to 0.6261 on extension bridge.

2) Hop-Conditioned Iteration at $k = 5$: Comparing ungated iterative retrieval at $k = 5$ against static fixed- $k = 5$ cross-encoder retrieval evaluates query reformula-

tion at greater depth. On the initial $N = 540$ evaluation split (bridge $N = 424$), iteration yields a modest recall difference from 0.8314 to 0.8514 ($\Delta = +0.0200$, 95% CI $[+0.0012, +0.0389]$, bootstrap raw $p = 0.0464$, Holm $p = 0.4640$; full-match McNemar unadjusted $p = 0.0187$, Holm $p = 0.1866$). On the $N = 2100$ extension split (bridge $N = 1653$), bridge recall shifts from 0.8445 to 0.8542 ($\Delta = +0.0097$, 95% CI $[0.0000, +0.0194]$, bootstrap raw $p = 0.0548$, Holm $p = 0.2192$; full-match McNemar unadjusted $p = 0.0344$, Holm $p = 0.1377$). On comparison questions, iteration at $k = 5$ yields lower recall than fixed- $k = 5$ ($0.9698 \rightarrow 0.9526$, $\Delta = -0.0172$, 95% CI $[-0.0431, +0.0043]$, bootstrap raw $p = 0.1974$, Holm

TABLE III
Full Retrieval Performance Across Evaluation Splits ($N = 540$ Eval and $N = 2100$ Extension)

Method	Type	Initial Eval Split ($N = 540$)				Extension Split ($N = 2100$)			
		Recall	Precision	Full-Match	Mean k	Recall	Precision	Full-Match	Mean k
Arm 1 (Dense Baseline $k = 4$)	Bridge	0.7559	0.3779	0.5354	4.000	0.7429	0.3728	0.5275	4.000
	Comparison	0.9138	0.4569	0.8362	4.000	0.9306	0.4691	0.8613	4.000
Arm 2 (Fixed- k CE, $k = 4$)	Bridge	0.8054	0.4027	0.6132	4.000	0.8076	0.4051	0.6261	3.995
	Comparison	0.9483	0.4741	0.8966	4.000	0.9553	0.4814	0.9128	3.984
Arm 2 (Fixed- k CE, $k = 5$)	Bridge	0.8314	0.3328	0.6627	4.998	0.8445	0.3394	0.6975	4.992
	Comparison	0.9698	0.3879	0.9397	5.000	0.9720	0.3939	0.9441	4.971
Arm 3 (Uncalib. Gating, $s \geq -1.925$)	Bridge	0.7818	0.4895	0.5684	4.158	—	—	—	—
	Comparison	0.9612	0.4983	0.9224	4.534	—	—	—	—
Arm 4a (Calib. Gating, $\theta^* = 0.18$)	Bridge	0.7394	0.5670	0.4858	3.535	—	—	—	—
	Comparison	0.9353	0.6046	0.8793	3.845	—	—	—	—
Arm 4b (Calib. Gating, $\theta^* = 0.18$)	Bridge	0.7417	0.5581	0.4906	3.594	0.7577	0.5659	0.5227	3.598
	Comparison	0.9224	0.6345	0.8534	3.629	0.9128	0.6673	0.8277	3.398
Arm 4b (Sensitivity, $\theta = 0.15$)	Bridge	0.7677	0.5113	0.5401	3.976	0.7834	0.5148	0.5729	3.998
	Comparison	0.9353	0.6046	0.8793	3.845	0.9295	0.6358	0.8613	3.640
Arm 5 (Iter. Gated, $\theta^* = 0.18$)	Bridge	0.7983	0.5107	0.6108	3.934	0.7895	0.5204	0.5935	3.850
	Comparison	0.8793	0.4947	0.7759	4.198	0.9295	0.5559	0.8591	4.038
Arm 5 (Sensitivity, $\theta = 0.15$)	Bridge	0.8243	0.4523	0.6604	4.455	0.8173	0.4582	0.6461	4.399
	Comparison	0.9138	0.4701	0.8362	4.560	0.9407	0.5333	0.8814	4.237
Iterative Ungated ($k = 4$)	Bridge	0.8160	0.4080	0.6462	4.000	0.8261	0.4143	0.6673	3.995
	Comparison	0.9181	0.4591	0.8534	4.000	0.9329	0.4702	0.8658	3.984
Iterative Ungated ($k = 5$)	Bridge	0.8514	0.3408	0.7099	4.998	0.8542	0.3433	0.7181	4.992
	Comparison	0.9526	0.3810	0.9052	5.000	0.9620	0.3899	0.9239	4.971
Arm 6 (Sufficiency Stopper, $\tau^* = 0.95$)	Bridge	0.7983	0.5107	0.6108	3.934	0.7895	0.5204	0.5935	3.850
	Comparison	0.8793	0.4947	0.7759	4.198	0.9295	0.5559	0.8591	4.038
Oracle Hybrid Upper Bound	Bridge	0.8160	0.4080	0.6462	4.000	0.8261	0.4143	0.6673	3.995
	Comparison	0.9483	0.4741	0.8966	4.000	0.9553	0.4814	0.9128	3.984

Notes on Table III: (1) The uncalibrated score gating baseline (Arm 3) and pooled Platt gating (Arm 4a) were evaluated on the initial $N = 540$ evaluation split as diagnostic comparators and were not evaluated on the $N = 2100$ extension cohort. (2) At $\tau^* = 0.95$, the sufficiency stopper never fires before the candidate score falls below $\theta^* = 0.18$, so the Arm 6 rows equal Arm 5 across both splits. (3) Arm 4a ($\theta^* = 0.18$) and Arm 4b ($\theta = 0.15$) give identical comparison rows on the $N = 540$ evaluation split because their raw-score cutoffs (-0.564 vs. -0.568 , verified from calibration parameters) select the same candidate chunks.

$p = 0.7896$ on initial eval; $0.9720 \rightarrow 0.9620$, $\Delta = -0.0101$, 95% CI $[-0.0235, +0.0034]$, bootstrap raw $p = 0.1646$, Holm $p = 0.3852$ on extension). Across both splits and question categories, these modest shifts at $k = 5$ fall short of statistical significance under Holm-Bonferroni correction.

3) Oracle Hybrid Upper Bound: By pairing iterative ungated retrieval ($k = 4$) on bridge queries with static fixed- k ($k = 4$) on comparison queries—using the ground-truth question-type label as an oracle router—the hybrid system attains 0.8240 bridge recall and 0.9538 comparison recall at a uniform depth of $k = 4.000$. This serves as an oracle upper bound, true by construction.

E. Hypothesis Testing at Reference Depth $k = 4.00$

Table IV details the formal hypothesis tests across 10,000 paired bootstrap resamples at the reference depth $k = 4.00$. In each resample, adaptive methods are re-interpolated to exactly $k = 4.00$, preserving the joint distribution over examples. P-values are adjusted via the step-down Holm-Bonferroni procedure to control family-

wise error at $\alpha = 0.05$ (separately for the 5 bridge contrasts and 3 comparison contrasts within each cohort).

- Arm 5 vs. Fixed- k on Bridge: With an observed delta of -0.0088 and 95% CI $[-0.0187, +0.0010]$, the raw bootstrap p -value is 0.0794 (Holm $p = 0.0794$). Consequently, we fail to reject the null hypothesis; iterative gated retrieval performs on par with fixed- k retrieval (the 95% CI includes zero).
- Gating Penalty on Bridge: Comparing Arm 5 directly to ungated iterative retrieval at matched depth reveals a significant recall penalty ($\Delta = -0.0257$, 95% CI $[-0.0328, -0.0186]$, raw $p = 0.0001$, Holm $p = 0.0005$). Confidence gating eliminates supporting paragraphs that fall below threshold.
- Single-Pass Calibrated Gating: Arm 4b is significantly inferior to fixed- k on bridge ($\Delta = -0.0266$, 95% CI $[-0.0341, -0.0191]$, Holm $p = 0.0005$), but is statistically indistinguishable from fixed- k on comparison ($\Delta = -0.0018$, 95% CI $[-0.0136, +0.0095]$, raw $p = 0.7202$, Holm $p = 0.7202$).
- Iterative vs. Single-Pass Gating: Arm 5 significantly

TABLE IV
Paired Bootstrap Hypothesis Testing ($B = 10,000$ Resamples) and Step-Down Holm-Bonferroni Correction at Reference Depth $k = 4.00$
(Family: 5 Bridge Contrasts and 3 Comparison Contrasts per Cohort)

Cohort	Hypothesis Comparator	Contrast Description	Δ Recall	95% Bootstrap CI	Raw p	Holm p	Verdict
Pooled Bridge ($N = 2077$)	Iter $k = 4$ vs. Arm 2 ($k = 4$)	Hop-conditioning alone at exact $k = 4.00$	+0.0169	[+0.0077, +0.0258]	< 0.001	0.001	Significant Gain
	Arm 5 (interp) vs. Iter $k = 4$	Recall cost of calibrated gating	-0.0257	[-0.0328, -0.0186]	< 0.001	< 0.001	Significant Cost
	Arm 5 (interp) vs. Arm 2 ($k = 4$)	Iterative gated vs. fixed- k at $k = 4.00$	-0.0088	[-0.0187, +0.0010]	0.079	0.079	No Detectable Diff. (CI includes 0)
	Arm 4b (interp) vs. Arm 2 ($k = 4$)	Single-pass gating vs. fixed- k at $k = 4.00$	-0.0266	[-0.0341, -0.0191]	< 0.001	< 0.001	Significant Loss
	Arm 5 (interp) vs. Arm 4b (interp)	Iterative vs. single-pass gating	+0.0177	[+0.0084, +0.0269]	< 0.001	0.001	Significant Gain
Pooled Comp ($N = 563$)	Iter $k = 4$ vs. Arm 2 ($k = 4$)	Hop-conditioning on comparison	-0.0240	[-0.0391, -0.0089]	0.002	0.007	Significant Loss
	Arm 4b (interp) vs. Arm 2 ($k = 4$)	Single-pass gating on comparison	-0.0018	[-0.0136, +0.0095]	0.720	0.720	Indistinguishable
	Arm 5 (interp) vs. Arm 2 ($k = 4$)	Iterative gating on comparison	-0.0391	[-0.0565, -0.0221]	< 0.001	< 0.001	Significant Loss

Holm-Bonferroni Family Definition: Statistical correction is applied separately to the family of 5 contrasts on the pooled bridge cohort and the family of 3 contrasts on the pooled comparison cohort, using step-down Holm adjusted significance thresholds at family-wise error rate $\alpha = 0.05$. Bootstrap p -values smaller than 0.001 are reported as $p < 0.001$.

outperforms Arm 4b on bridge ($\Delta = +0.0177$, 95% CI [+0.0084, +0.0269], raw $p = 0.0002$, Holm $p = 0.0006$).

F. Exploratory Multi-Depth Matched Analysis ($k \in [2.0, 5.5]$)

To resolve whether adaptive gating exhibits depth-dependent trade-offs beyond the single $k = 4.00$ operating point, Table V reports an exploratory matched-depth analysis evaluating target depths $k \in \{2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5\}$. We interpolate each method per bootstrap resample ($B = 10,000$) and apply step-down Holm-Bonferroni adjustment across the 8-depth family per contrast.

1) Depth-Dependent Trajectory of Arm 5: On bridge questions, Arm 5's relative recall varies across retrieval depths:

- Low Retrieval Budgets ($k \leq 3.5$): Iterative gating suffers a statistically significant recall deficit relative to static fixed- k ($\Delta = -0.0522$ at $k = 2.0$; $\Delta = -0.0376$ at $k = 2.5$; $\Delta = -0.0338$ at $k = 3.0$; $\Delta = -0.0178$ at $k = 3.5$; all Holm $p \leq 0.005$). In this regime, score thresholding terminates multi-hop retrieval paths prematurely.
- Intermediate Retrieval Budgets ($4.0 \leq k \leq 5.0$): Arm 5 converges with fixed- k , remaining statistically indistinguishable from fixed- k at $k = 4.0$ ($\Delta = -0.0088$, Holm $p = 0.2382$, 95% CI includes zero), $k = 4.5$ ($\Delta = -0.0016$, Holm $p = 1.0000$), and $k = 5.0$ ($\Delta = +0.0031$, Holm $p = 1.0000$).
- Deep Retrieval Budgets ($k = 5.5$): At $k = 5.5$, Arm 5 attains +0.0127 recall over fixed- k (95% CI [+0.0044, +0.0210], raw $p = 0.0044$, Holm $p = 0.0176$). We label this an exploratory gain, as the depth grid was selected post-hoc.

2) Arm 6: Early Stopping vs. Fixed- k Comparison: The multi-depth grid provides clarity on the learned sufficiency stopper (Arm 6):

- Versus Fixed- k on Bridge: On bridge questions, Arm 6 achieves recall comparable to fixed- k at $k = 2.0$ (0.6814 vs. 0.6810, $\Delta = +0.0003$), $k = 3.0$ (0.7616 vs.

0.7617, $\Delta = -0.0000$), and $k = 3.5$ (0.7811 vs. 0.7844, $\Delta = -0.0033$). At $k = 2.5$, Arm 6 attains a nominal recall difference of +0.0088 (0.7302 vs. 0.7214); however, its 95% bootstrap CI [-0.0008, +0.0183] crosses zero (unadjusted $p = 0.0720$, Holm $p = 0.2880$). Thus, Arm 6 performs on par with fixed- k on bridge questions across all covered depths ($k \in [2.0, 3.5]$, consistent with no difference as all 95% CIs span zero).

- Versus Fixed- k on Comparison: On comparison questions, Arm 6 is significantly worse than fixed- k at every covered depth $k \in [2.0, 3.5]$, with deficits of $\Delta = -0.1376$ at $k = 2.0$, $\Delta = -0.0901$ at $k = 2.5$, $\Delta = -0.0659$ at $k = 3.0$, and $\Delta = -0.0441$ at $k = 3.5$ (all Holm $p < 0.001$).
- Versus Arm 5 on Both Types: In exploratory analysis, Arm 6 provides statistically significant recall improvements over iterative gating (Arm 5) on both question types throughout the low-depth regime $k \in [2.0, 3.5]$: +0.0525 at $k = 2.0$, +0.0464 at $k = 2.5$, +0.0338 at $k = 3.0$, and +0.0145 at $k = 3.5$ on bridge (all Holm $p < 0.001$); and +0.0410 at $k = 2.0$, +0.0547 at $k = 2.5$, +0.0539 at $k = 3.0$, and +0.0344 at $k = 3.5$ on comparison (all Holm $p < 0.001$). We hypothesize that, at equal mean depth, the sufficiency stopper spends retrieval depth more effectively than raising θ because it stops when the selected set already appears sufficient.
- Coverage Limit: Under frozen calibration threshold $\theta^* = 0.18$, Arm 6 plateaus at mean depth $k = 3.87$ on bridge, so it never reaches depth $k = 4.00$; on comparison questions, Arm 6 reaches mean depths up to $k \approx 4.04$ –4.20 (observed point estimate 0.9168 at $k = 4.0$), but fewer than 95% of bootstrap resamples cover depth $k = 4.00$, precluding valid bootstrap hypothesis testing and confidence intervals. Depths $k \in \{4.0, 4.5, 5.0, 5.5\}$ are thus classified as Not Covered. Furthermore, Arm 6 relies on the oracle question-type indicator (is_bridge) and calibration-split hyperparameter tuning.

TABLE V

Exploratory Multi-Depth Matched-Depth Analysis (Selected Comparisons Across $k \in \{2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5\}$, Pooled $N = 2640$; All Listed Rows Satisfy $\geq 95\%$ Bootstrap Coverage; Holm Family: 8 Depths per Contrast)

Cohort	Comparison	Depth k	Δ Recall	95% Bootstrap CI	Raw p	Holm p	Verdict
Bridge ($N = 2077$)	Arm 6 vs. Fixed- k	$k = 2.0$	+0.0003	$[-0.0100, +0.0105]$	0.9524	1.0000	No Detectable Diff.
		$k = 2.5$	+0.0088	$[-0.0008, +0.0183]$	0.0720	0.2880	No Detectable Diff.
		$k = 3.0$	-0.0000	$[-0.0099, +0.0102]$	0.9866	1.0000	No Detectable Diff.
		$k = 3.5$	-0.0033	$[-0.0132, +0.0067]$	0.5170	1.0000	No Detectable Diff.
Bridge ($N = 2077$)	Arm 6 vs. Arm 5	$k = 2.0$	+0.0525	$[+0.0453, +0.0596]$	< 0.001	< 0.001	Significant Gain
		$k = 2.5$	+0.0464	$[+0.0390, +0.0538]$	< 0.001	< 0.001	Significant Gain
		$k = 3.0$	+0.0338	$[+0.0272, +0.0402]$	< 0.001	< 0.001	Significant Gain
		$k = 3.5$	+0.0145	$[+0.0095, +0.0198]$	< 0.001	< 0.001	Significant Gain
Bridge ($N = 2077$)	Arm 5 vs. Fixed- k	$k = 2.0$	-0.0522	$[-0.0626, -0.0418]$	< 0.001	< 0.001	Significant Loss
		$k = 2.5$	-0.0376	$[-0.0474, -0.0276]$	< 0.001	< 0.001	Significant Loss
		$k = 3.0$	-0.0338	$[-0.0442, -0.0232]$	< 0.001	< 0.001	Significant Loss
		$k = 3.5$	-0.0178	$[-0.0276, -0.0079]$	0.0010	0.0050	Significant Loss
		$k = 4.0$	-0.0088	$[-0.0187, +0.0010]$	0.0794	0.2382	No Detectable Diff.
		$k = 4.5$	-0.0016	$[-0.0107, +0.0074]$	0.7344	1.0000	No Detectable Diff.
		$k = 5.0$	+0.0031	$[-0.0058, +0.0120]$	0.5100	1.0000	No Detectable Diff.
		$k = 5.5$	+0.0127	$[+0.0044, +0.0210]$	0.0044	0.0176	Exploratory Gain
Comparison ($N = 563$)	Arm 6 vs. Fixed- k	$k = 2.0$	-0.1376	$[-0.1576, -0.1175]$	< 0.001	< 0.001	Significant Loss
		$k = 2.5$	-0.0901	$[-0.1094, -0.0702]$	< 0.001	< 0.001	Significant Loss
		$k = 3.0$	-0.0659	$[-0.0861, -0.0453]$	< 0.001	< 0.001	Significant Loss
		$k = 3.5$	-0.0441	$[-0.0617, -0.0260]$	< 0.001	< 0.001	Significant Loss
Comparison ($N = 563$)	Arm 6 vs. Arm 5	$k = 2.0$	+0.0410	$[+0.0301, +0.0521]$	< 0.001	< 0.001	Significant Gain
		$k = 2.5$	+0.0547	$[+0.0420, +0.0673]$	< 0.001	< 0.001	Significant Gain
		$k = 3.0$	+0.0539	$[+0.0413, +0.0669]$	< 0.001	< 0.001	Significant Gain
		$k = 3.5$	+0.0344	$[+0.0207, +0.0491]$	< 0.001	< 0.001	Significant Gain
Comparison ($N = 563$)	Arm 5 vs. Fixed- k	$k = 2.0$	-0.1786	$[-0.1978, -0.1592]$	< 0.001	< 0.001	Significant Loss
		$k = 2.5$	-0.1447	$[-0.1632, -0.1259]$	< 0.001	< 0.001	Significant Loss
		$k = 3.0$	-0.1198	$[-0.1397, -0.0992]$	< 0.001	< 0.001	Significant Loss
		$k = 3.5$	-0.0785	$[-0.0974, -0.0593]$	< 0.001	< 0.001	Significant Loss
		$k = 4.0$	-0.0391	$[-0.0565, -0.0221]$	< 0.001	< 0.001	Significant Loss
		$k = 4.5$	-0.0183	$[-0.0329, -0.0043]$	0.0110	0.0330	Significant Loss
		$k = 5.0$	-0.0058	$[-0.0181, +0.0060]$	0.3470	0.6940	No Detectable Diff.

Exploratory Analysis Disclaimer and Holm Family: This multi-depth matched analysis is designated as exploratory because the 8-depth evaluation grid $k \in \{2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5\}$ was chosen post-hoc following empirical inspection of the Pareto curves. Step-down Holm-Bonferroni correction is applied across the 8-depth family per contrast. For Arm 6 vs. fixed- k , performance indistinguishable from fixed- k holds strictly on bridge questions; on comparison questions, Arm 6 is significantly worse than fixed- k at every covered depth $k \in [2.0, 3.5]$ ($\Delta = -0.138$ to -0.044 , all Holm $p < 0.001$). Arm 6 significantly improves upon Arm 5 on both question types across $k \in [2.0, 3.5]$ (all Holm $p < 0.001$). All listed comparisons satisfy the $\geq 95\%$ bootstrap resample coverage criterion; depths where fewer than 95% of paired resamples cover the target depth without extrapolation are designated Not Covered. Specifically, on bridge questions, Arm 6 plateaus at mean depth $k = 3.87$ under frozen calibration threshold $\theta^* = 0.18$ and never reaches depths $k \in \{4.0, 4.5, 5.0, 5.5\}$. On comparison questions, Arm 6 reaches mean depths of $k \approx 4.04$ – 4.20 (observed point estimate 0.9168 at $k = 4.0$), but fewer than 95% of paired bootstrap resamples cover depth $k = 4.00$, so depths $k \geq 4.0$ are classified as Not Covered for hypothesis testing.

G. Ablation: Full-Chunk vs. 50-Word Query Augmentation in Arm 5

To evaluate whether appending the full previously selected chunk introduces query drift or token dilution during later hops of cross-encoder re-ranking, we compare full-chunk augmentation (default Arm 5) against a 50-word prefix truncation variant on the $N = 540$ evaluation split across both operating thresholds ($\theta^* = 0.18$ and sensitivity $\theta = 0.15$). We note that Arm 5 is iterative and hop-conditioned, concatenating previously selected chunks into queries for later hops, whereas Arm 4b is single-pass and has no query augmentation. This ablation is a single-split descriptive comparison with no significance test:

- At Operating Threshold $\theta^* = 0.18$: Full-chunk aug-

mentation achieves bridge recall 0.7983 ($k = 3.934$, precision 0.5107, full-match 0.6108) and comparison recall 0.8793 ($k = 4.198$, precision 0.4947, full-match 0.7759). Truncating to a 50-word prefix yields bridge recall 0.8031 ($k = 4.033$, precision 0.4983, full-match 0.6179) and comparison recall 0.9009 ($k = 4.328$, precision 0.4864, full-match 0.8190).

- At Sensitivity Threshold $\theta = 0.15$: Full-chunk augmentation attains bridge recall 0.8243 ($k = 4.455$, precision 0.4523, full-match 0.6604) and comparison recall 0.9138 ($k = 4.560$, precision 0.4701, full-match 0.8362). Truncating to 50 words yields identical bridge recall of 0.8243 ($k = 4.493$, precision 0.4494, full-match 0.6580) and comparison recall 0.9353 ($k =$

4.595, precision 0.4691, full-match 0.8793).

The observed differences between 50-word prefix truncation and full-chunk augmentation in Arm 5 are: bridge recall is +0.005 at $\theta^* = 0.18$ and 0.000 at $\theta = 0.15$; comparison recall is +0.022 at $\theta^* = 0.18$ and +0.022 at $\theta = 0.15$, with k also higher by ~ 0.1 chunks in each case.

V. Discussion

Our empirical findings reveal that neither post-hoc calibration nor confidence-based gating enables adaptive- k retrieval to consistently surpass static fixed-depth baselines in multi-hop RAG. We offer three mechanistic hypotheses to interpret these observations.

A. Hypothesis 1: Chunk-Level Calibration vs. Set-Level Sufficiency

We hypothesize that the core bottleneck of calibrated gating stems from the tension between chunk-level relevance calibration and set-level evidence sufficiency. Platt scaling is designed to ensure that among candidate chunks assigned predicted probability $\hat{p} \approx 0.30$, approximately 30% constitute genuine supporting documents in distribution—an alignment verified empirically, reducing ECE to 0.0391 on evaluation and 0.0373 on extension. Yet resolving multi-hop questions requires assembling disjoint supporting facts jointly. An independent stopping rule that drops candidates when individual probability falls below θ prematurely severs multi-hop chains: bridge passages often share less initial lexical overlap with the query than first-hop texts. Consequently, chunk-level thresholding prunes reasoning chains mid-traversal, degrading set-level recall at tight budgets.

B. Hypothesis 2: Asymmetric Query Reformulation Dynamics

The polarized impact of hop-conditioned query augmentation—advancing bridge questions (+1.7 points, $\Delta = +0.017$) while impairing comparison queries (−2.4 points, $\Delta = -0.024$)—stems directly from the underlying graph topology of the reasoning task:

- **Bridge Questions:** Bridge queries follow a sequential chain ($A \rightarrow B \rightarrow C$). The prompt mentions entity A , while evidence regarding C requires uncovering intermediate link entity B . Concatenating the text of the previously retrieved chunk injects entity B into the cross-encoder input, bridging the semantic gap.
- **Comparison Questions (Hypothesized Query Drift):** Comparison queries follow a parallel structure (A vs. B), explicitly mentioning both target entities in the initial question (e.g., “Were Scott Derrickson and Ed Wood of the same nationality?”). We hypothesize that appending context describing entity A over-indexes cross-encoder attention onto A , crowding out entity B during subsequent re-ranking steps. This hypothesized query drift impairs comparison recall.

C. Hypothesis 3: Oracle Routing as an Upper Bound

Our assessment of the Oracle Hybrid pipeline demonstrates that optimal retrieval policy is intrinsically tied to question topology. While score-level thresholding fails

to reliably beat fixed- k , routing queries according to reasoning topology achieves 0.8240 bridge recall and 0.9538 comparison recall at $k = 4.00$. However, because this upper bound presumes access to ground-truth question classifications, practical deployments must prioritize topology-detection classifiers prior to context assembly.

VI. Threats to Validity and Limitations

We outline eight primary limitations governing our study:

- 1) **Benchmark Generalization:** Our investigation is grounded within HotpotQA [5]. Although HotpotQA represents a canonical multi-hop benchmark, alternate datasets (such as 2WikiMultiHopQA and MuSiQue) present differing reasoning hop depths, negative distractor distributions, and entity graph topologies.
- 2) **Distractor Setting Scope:** All experiments operate in HotpotQA’s distractor setting, evaluating a per-query pool of ~ 10 passages (2 ground-truth facts, 8 challenging distractors). While this protocol isolates re-ranking precision and threshold behavior against hard negatives, it does not assess full-scale open-domain retrieval across web-scale corpora.
- 3) **Focus on Retrieval Metrics:** Our evaluations focus on retrieval-stage fidelity—specifically passage recall, precision, and full-match rate. While retrieval recall bounds context availability for generation, we do not assess downstream language model synthesis metrics (e.g., Exact Match or token F1).
- 4) **Model Architectures:** We investigated a single bi-encoder (all-MiniLM-L6-v2) and a single cross-encoder (ms-marco-MiniLM-L-6-v2). Larger model scales (e.g., 7B-parameter cross-encoders or generative re-rankers) may display distinct calibration characteristics and representational power.
- 5) **Context Window Truncation:** Cross-encoder inputs were capped at 512 tokens, which may occasionally truncate passage content during concatenated multi-hop re-ranking.
- 6) **Reliance on Oracle Question Types:** The hybrid upper bound and Arm 6’s `is_bridge` feature depend on ground-truth question-type metadata. Real-world systems must predict question topology dynamically.
- 7) **Hyperparameter Selection Split:** Decision cutoffs ($\theta^* = 0.18$, $\tau^* = 0.95$, $s_{\text{thresh}} = -1.9250$) were established using the $N = 360$ calibration subset. Alternate data distributions may favor different decision thresholds.
- 8) **Exploratory Grid Selection and Arm 6 Coverage:** The multi-depth grid was chosen post-hoc after inspecting empirical Pareto curves, and depth $k = 5.5$ sits at the parameter sweep margin where bootstrap coverage is constrained; thus, the gain (+0.0127, Holm $p = 0.0176$) is exploratory. Additionally, on bridge questions Arm 6 plateaus at mean depth

$k = 3.87$ and never reaches $k = 4.00$; on comparison questions, point estimates reach $k \approx 4.04$ – 4.20 , but fewer than 95% of resamples cover $k = 4.00$, precluding non-parametric hypothesis testing.

A. Reproducibility

Experimental pipelines used Python 3.14, PyTorch 2.12.1, Hugging Face transformers 5.12.1, sentence-transformers 6.1.0, scikit-learn 1.8.0, and datasets 5.0.0. From the 7,405 validation items in HotpotQA’s distractor setting, we applied seed-42 shuffling and partitioned the initial 3,000 instances into calibration ($N = 360$), evaluation ($N = 540$), and extension ($N = 2100$) splits. All calibration parameters (Platt coefficients, $\theta^* = 0.18$, $\tau^* = 0.95$, $s_{\text{thresh}} = -1.9250$) were fit once on the calibration split and held frozen across all evaluation experiments.

VII. Conclusion

This study evaluated post-hoc confidence calibration, hop-conditioned query reformulation, and adaptive retrieval depth in multi-hop RAG. Although Platt scaling reduces cross-encoder calibration error by over 80%, gating retrieval decisions on calibrated probabilities fails to outperform static fixed- k baselines at matched depth $k = 4.00$. Furthermore, confidence-based gating imposes a noticeable recall penalty relative to ungated multi-step iteration on bridge questions. In exploratory multi-depth analysis across $k \in [2.0, 5.5]$, iterative gating lags behind fixed- k at constrained budgets before matching fixed- k at intermediate depths; on bridge questions, a learned sufficiency stopper matches fixed- k across all covered depths (all 95% CIs include zero), but on comparison questions it is significantly worse than fixed- k at every covered depth $k \in [2.0, 3.5]$ ($\Delta = -0.138$ to -0.044 , all Holm $p < 0.001$); nonetheless, in exploratory evaluation it improves upon iterative gating (Arm 5) on both question types across $k \in [2.0, 3.5]$ by $+1.5$ to $+5.5$ points (all Holm $p < 0.001$). Sequential query expansion yields clear benefits for chained bridge questions while impairing parallel comparison queries. These findings show that adaptive retrieval depth cannot be resolved through passage-level score thresholding alone; future research in multi-hop RAG should focus on topology-guided query routing and set-level evidence sufficiency.

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